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2014.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the engine cooling system and operation, refer to the relevant Description and Operation section of the workshop manual. REFER to: Engine Cooling (303-03B, Description and Operation).

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.



NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Coolant leaks ▪ Coolant Hoses ▪ Coolant expansion tank ▪ Radiator ▪ Heater core ▪ Accessory drive belt ▪ Viscous fan	▪ Fuses ▪ Harnesses ▪ Loose or corroded connector(s) ▪ Engine Coolant Temperature (ECT) sensor

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Coolant loss	▪ Hoses ▪ Hose connections ▪ Radiator ▪ Water pump ▪ Heater core ▪ Gaskets ▪ Engine casting cracks ▪ Engine block core plugs	Carry out a visual inspection. If there are no obvious leaks, carry out a cooling system pressure test. Rectify any leaks as necessary.
Overheating	▪ Low /Contaminated coolant ▪ Thermostat ▪ Viscous fan ▪ ECT sensor ▪ Restricted air flow over the radiator	Check the coolant level and condition. Carry out a cooling system pressure test. Rectify any leaks as necessary. Check the thermostat and rectify as necessary. Check the viscous fan operation, make sure the viscous fan rotates freely. Check for obstructions to the air flow over the radiator. Rectify as necessary.
Engine not reaching normal temperature	▪ Thermostat ▪ Viscous fan ▪ Thermostat ▪ Electric fan ▪ Fan speed module	Check the thermostat operation. Check the viscous fan operation, make sure the viscous fan is not seized. Rectify as necessary.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: Diagnostic Trouble Code (DTC) Index - 5.0L, DTC: Module Name: Engine Control Module (100-00, Description and Operation).